Topic 13

Vibrations and Waves

CENGAGE

Reading Question 13-1 (1 of 2)

Suppose that a grandfather clock is calibrated correctly at sea level and is then brought up to the top of a tall mountain. What would need to be done to calibrate the clock at the top of the mountain?

1. Nothing, it should work fine since it was already calibrated at sea level.
2. The bob of the pendulum should be slid slightly down the rod.
3. The bob of the pendulum should be slid slightly up the rod.
4. It cannot be calibrated at the top of the mountain.

Reading Question 13-1 (2 of 2)

Suppose that a grandfather clock is calibrated correctly at sea level and is then brought up to the top of a tall mountain. What would need to be done to calibrate the clock at the top of the mountain?

1. Nothing, it should work fine since it was already calibrated at sea level.
2. The bob of the pendulum should be slid slightly down the rod.
3. **Answer: The bob of the pendulum should be slid slightly up the rod.**
4. It cannot be calibrated at the top of the mountain.

Reading Question 13-2 (1 of 2)

Which of these represents an example of simple harmonic motion?

(i) a bouncing ball

(ii) a marble rolling back and forth in a concave bowl

(iii) a tree branch vibrating

(iv) someone walking back and forth across a room,

1. all of them
2. (i) and (iv)
3. (ii) and (iii)
4. (i) and (iii)

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Reading Question 13-2 (2 of 2)

Which of these represents an example of simple harmonic motion?

(i) a bouncing ball

(ii) a marble rolling back and forth in a concave bowl

(iii) a tree branch vibrating

(iv) someone walking back and forth across a room,

1. all of them
2. (i) and (iv)
3. **Answer: (ii) and (iii)**
4. (i) and (iii)

Reading Question 13-3 (1 of 2)

For which of the following would it be most advisable to utilize a system that is damped?

1. within the pendulum of a grandfather clock
2. within shock absorbers
3. within the bouncing mechanism for a trampoline
4. within a buoy that floats on the ocean

Reading Question 13-3 (2 of 2)

For which of the following would it be most advisable to utilize a system that is damped?

1. within the pendulum of a grandfather clock
2. **Answer: within shock absorbers**
3. within the bouncing mechanism for a trampoline
4. within a buoy that floats on the ocean

Reading Question 13-4 (1 of 2)

Which of the following is true for an oscillating spring?

1. The kinetic energy is greatest when the spring is the most compressed and the most extended.
2. The potential energy is greatest when the spring is the most compressed and the most extended.
3. The total energy is greatest when the spring is the most compressed and the most extended.
4. None of these.

Reading Question 13-4 (2 of 2)

Which of the following is true for an oscillating spring?

1. The kinetic energy is greatest when the spring is the most compressed and the most extended.
2. **Answer: The potential energy is greatest when the spring is the most compressed and the most extended.**
3. The total energy is greatest when the spring is the most compressed and the most extended.
4. None of these.

Reading Question 13-5 (1 of 2)

Which of the following statements are true?

(i) When a pulse on a rope reflects from a rigid boundary, it is always inverted.

(ii) When a pulse on rope encounters a second rope (with different characteristics from the first) that is tied to the first one, there is only sometimes a reflected pulse.

1. They are both false.
2. (i) is true, but (ii) is false
3. (ii) is true, but (i) is false
4. They are both true.

Reading Question 13-5 (2 of 2)

Which of the following statements are true?

(i) When a pulse on a rope reflects from a rigid boundary, it is always inverted.

(ii) When a pulse on rope encounters a second rope (with different characteristics from the first) that is tied to the first one, there is only sometimes a reflected pulse.

1. They are both false.
2. **Answer: (i) is true, but (ii) is false**
3. (ii) is true, but (i) is false
4. They are both true.

Reading Question 13-6 (1 of 2)

Consider a rope hung from the top of a tall ceiling to the ground. Someone sets up waves from the bottom of the rope and watches as they propagate up the rope. The wave's speed along the vertical rope will

1. remain constant up the rope.
2. increase with vertical position up the rope.
3. decrease with vertical position up the rope.
4. increase until midway up the rope, then decrease.

Reading Question 13-6 (2 of 2)

Consider a rope hung from the top of a tall ceiling to the ground. Someone sets up waves from the bottom of the rope and watches as they propagate up the rope. The wave's speed along the vertical rope will

1. remain constant up the rope.
2. **Answer: increase with vertical position up the rope.**
3. decrease with vertical position up the rope.
4. increase until midway up the rope, then decrease.

Reading Question 13-7 (1 of 2)

For which of the following will the phenomenon of wave interference apply? (Choose the best answer.)

1. square-shaped pulses
2. sinusoidal pulses
3. triangular pulses
4. all of these

Reading Question 13-7 (2 of 2)

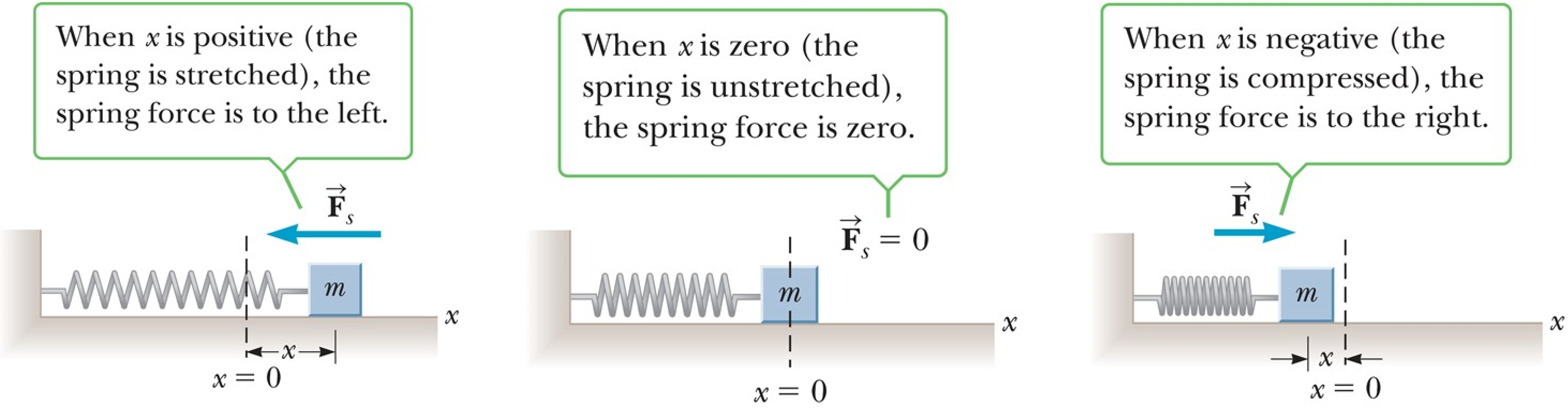
For which of the following will the phenomenon of wave interference apply? (Choose the best answer.)

1. square-shaped pulses
2. sinusoidal pulses
3. triangular pulses
4. **Answer: all of these**

Topic 13: Vibrations and Waves

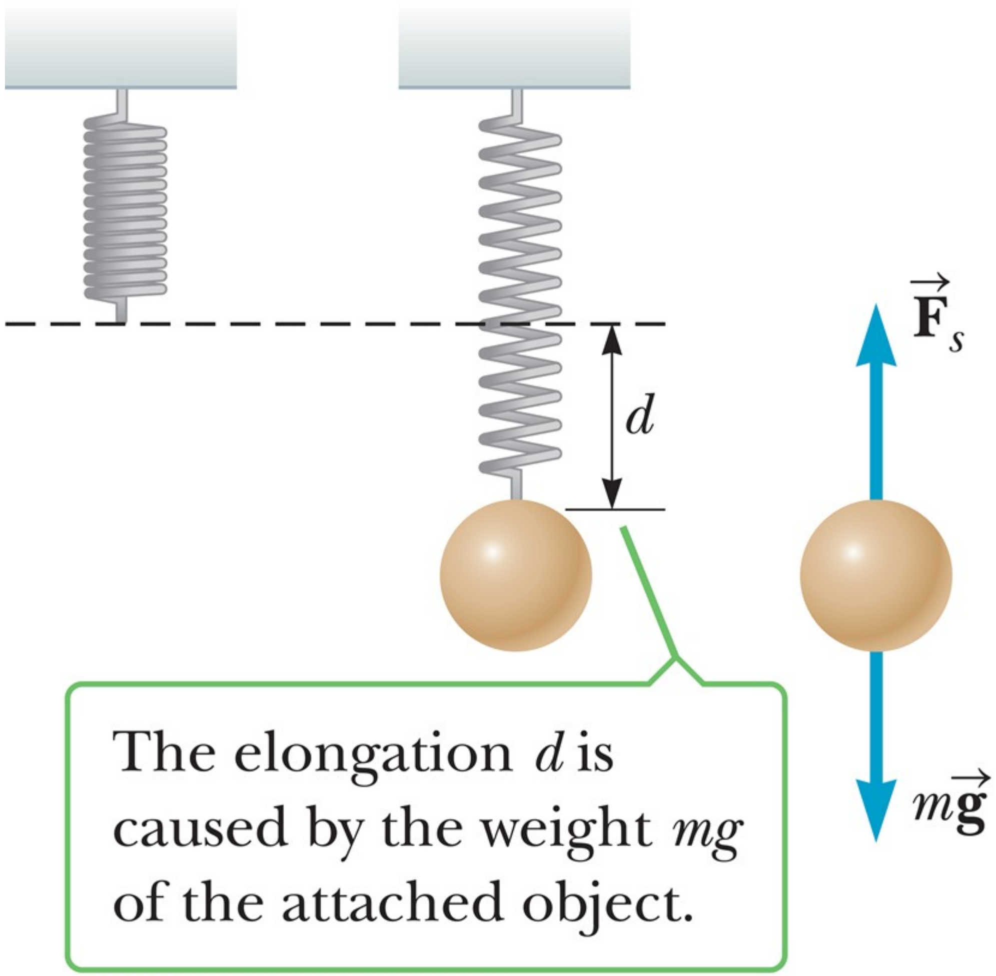
CHAPTER DISCUSSION

Hooke's Law (1 of 4)



$$F_s = -kx$$

Hooke’s Law (2 of 4)



Hooke's Law (3 of 4)

Amplitude (A): object oscillates between $x = -A$ and $x = +A$

Period (T): time four one cycle: from $x = A$ to $x = -A$ and back to $x = A$

Frequency (f): the number of complete cycles or vibrations per unit of time

$$f = \frac{1}{T}$$

Hooke's Law (4 of 4)

$$F = ma$$

$$-kx = ma \quad \rightarrow \quad a = -\frac{k}{m}x$$

$$-\frac{kA}{m} \leq a \leq \frac{kA}{m} \quad \rightarrow \quad a_{\max} = \frac{kA}{m}$$

Think – Pair – Share 1

A block on the end of a horizontal spring is pulled from equilibrium at $x = 0$ to $x = A$ and released. Through what total distance does it travel in one full cycle of its motion?

1. $A/2$
2. A
3. $2A$
4. $4A$

Think – Pair – Share 2

For a simple harmonic oscillator, which of the following pairs of vector quantities can't both point in the same direction? (The position vector is the displacement from equilibrium.)

1. position and velocity
2. velocity and acceleration
3. position and acceleration

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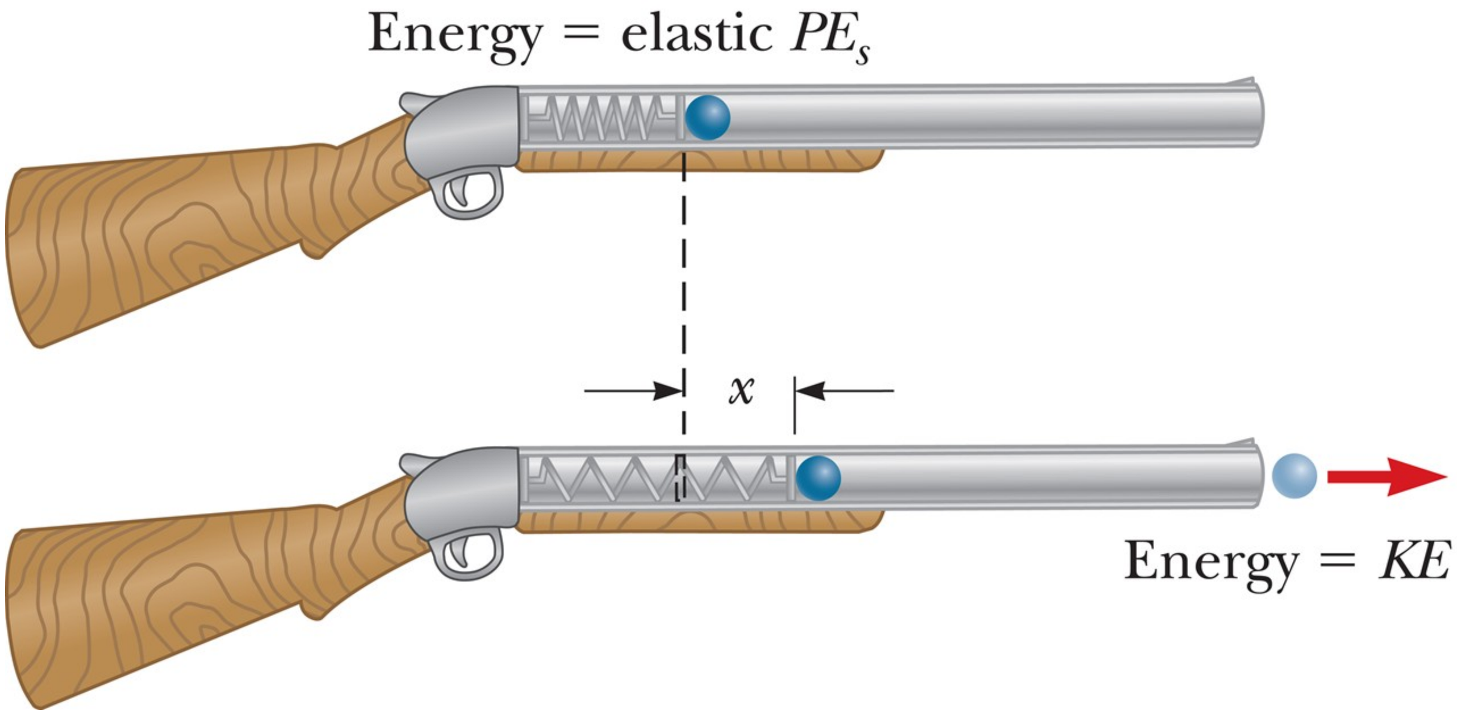
Elastic Potential Energy (1 of 7)



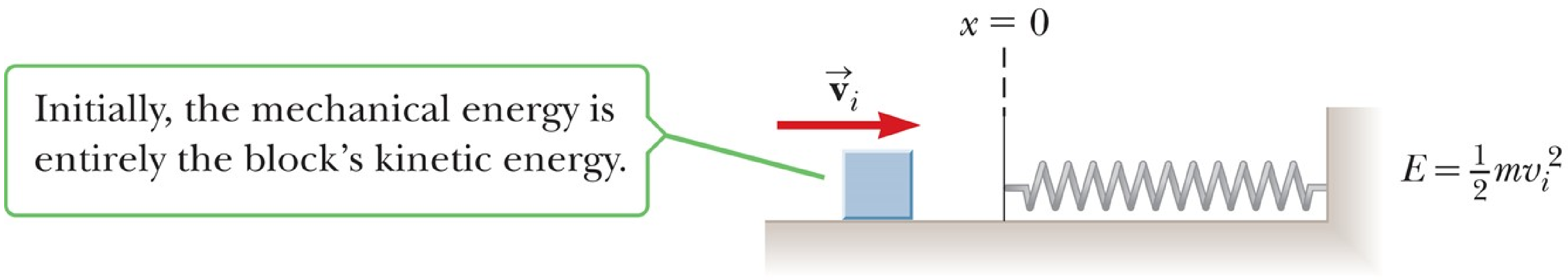
$$PE_s \equiv \frac{1}{2}kx^2$$

Elastic Potential Energy (2 of 7)

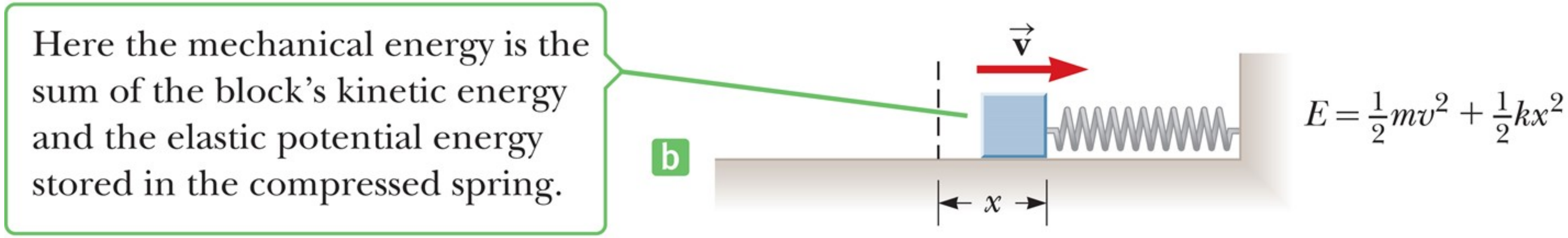
$$(KE + PE_g + PE_s)_i = (KE + PE_g + PE_s)_f$$

$$W_{nc} = (KE + PE_g + PE_s)_f - (KE + PE_g + PE_s)_i$$

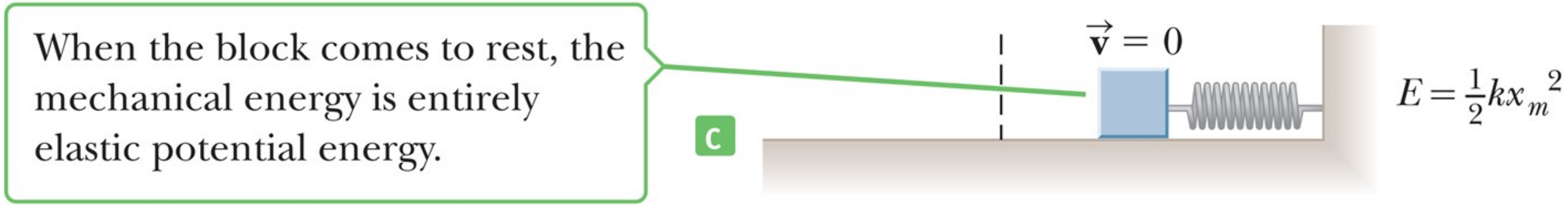
Elastic Potential Energy (3 of 7)



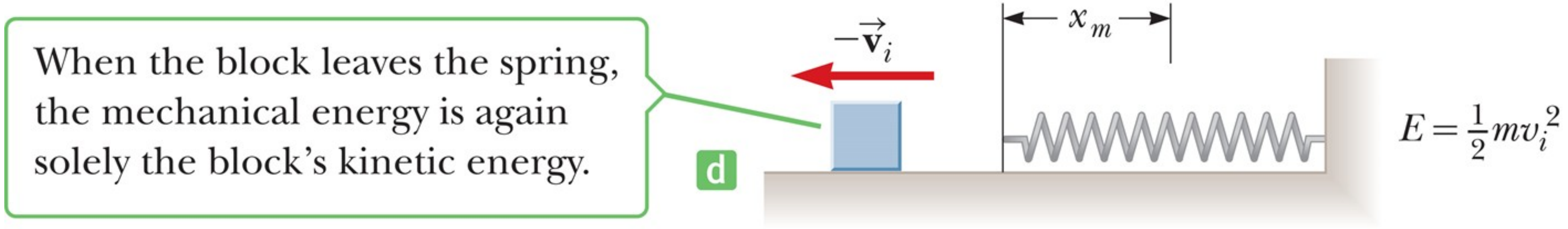
Elastic Potential Energy (4 of 7)



Elastic Potential Energy (5 of 7)



Elastic Potential Energy (6 of 7)



Elastic Potential Energy (7 of 7)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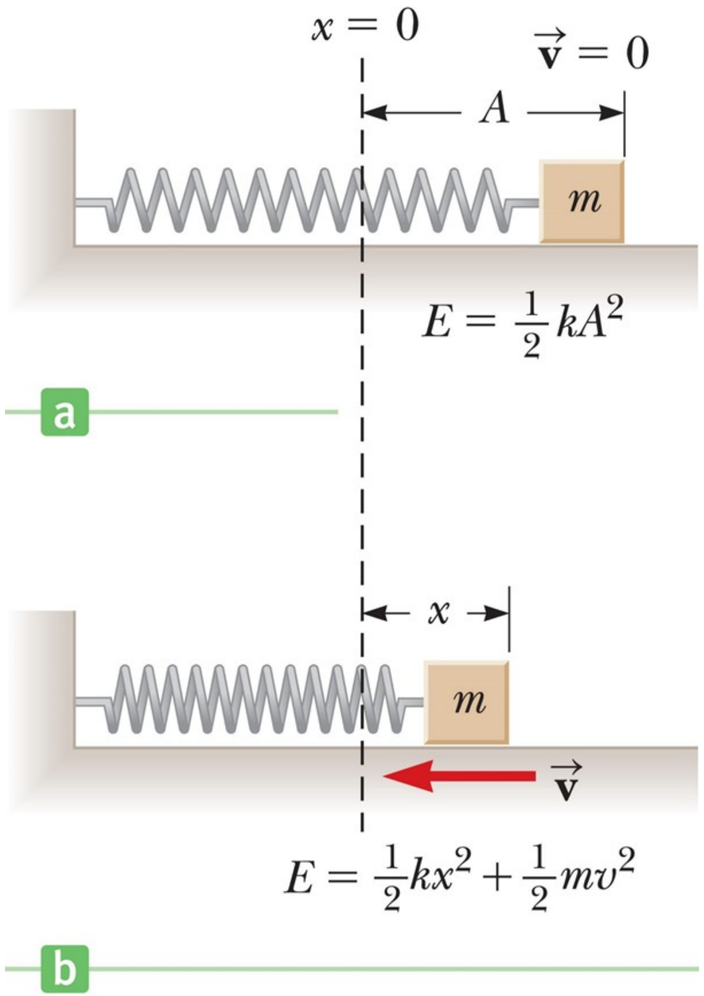
Think – Pair – Share 3

When an object moving in simple harmonic motion is at its maximum displacement from equilibrium, which of the following is at a maximum?

1. velocity
2. acceleration
3. kinetic energy

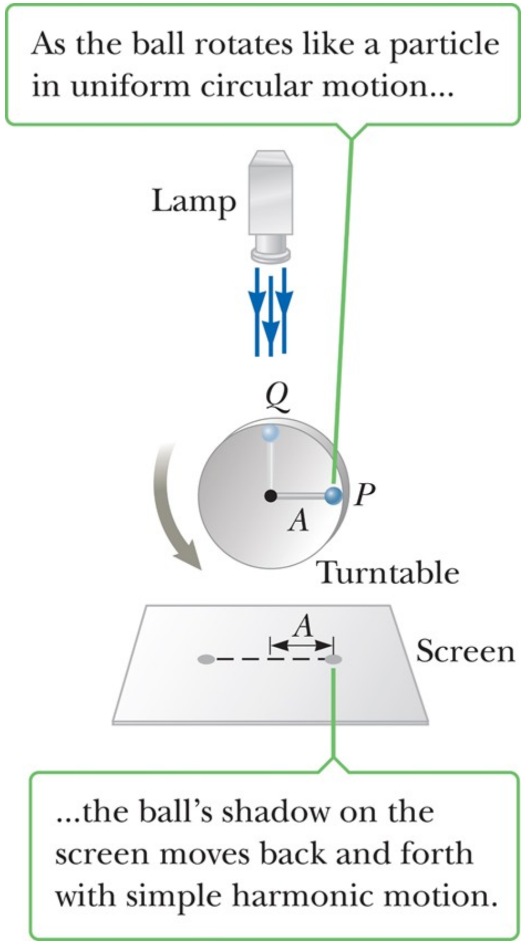
Velocity as a Function of Position



$$\tfrac{1}{2}kA^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}kx^2$$

$$v = \pm\sqrt{\frac{k}{m}\left(A^2 - x^2\right)}$$

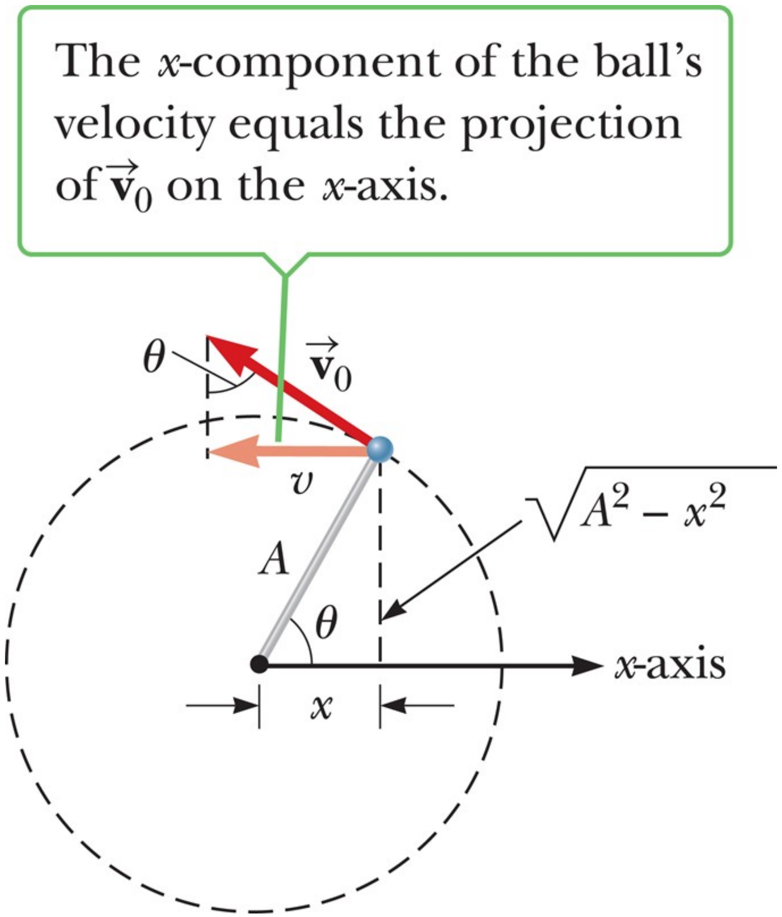
Comparing Simple Harmonic Motion with Uniform Circular Motion (1 of 3)



$$v = \pm\sqrt{\frac{k}{m}\left(A^2 - x^2\right)}$$

$$v = C\sqrt{A^2 - x^2}$$

Comparing Simple Harmonic Motion with Uniform Circular Motion (2 of 3)



$$v = v_0 \sin\theta \quad \rightarrow \quad \sin\theta = \frac{v}{v_0}$$

$$\sin\theta = \frac{\sqrt{A^2 - x^2}}{A}$$

$$\frac{v}{v_0} = \frac{\sqrt{A^2 - x^2}}{A}$$

$$v = \frac{v_0}{A}\sqrt{A^2 - x^2} = C\sqrt{A^2 - x^2}$$

Comparing Simple Harmonic Motion with Uniform Circular Motion (3 of 3)

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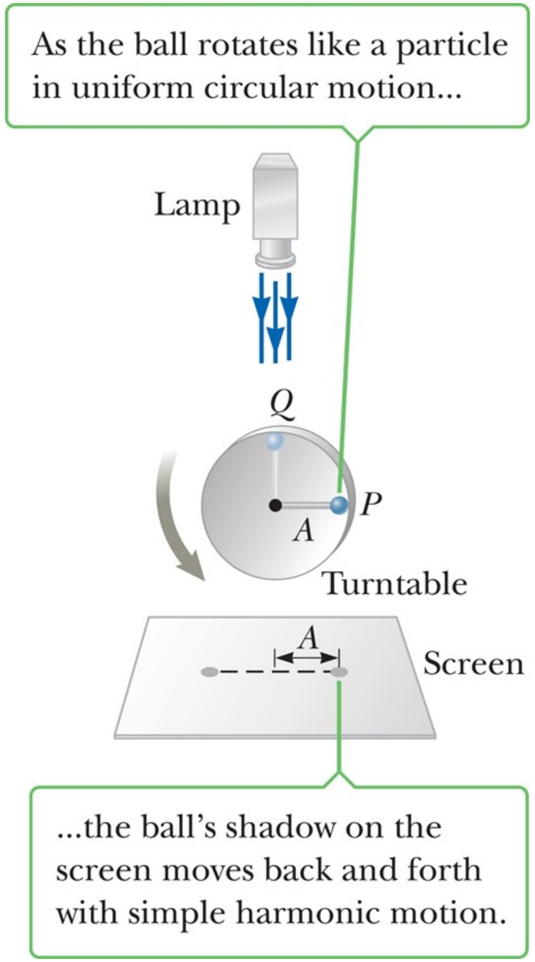
Period, Frequency, and Angular Frequency (1 of 2)



$$v_0 = \frac{2\pi A}{T} \qquad T = \frac{2\pi A}{v_0}$$

$$\tfrac{1}{2}kA^2 = \tfrac{1}{2}mv_0^2$$

$$\frac{A}{v_0} = \sqrt{\frac{m}{k}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Period, Frequency, and Angular Frequency (2 of 2)

$$f = \frac{1}{T} \quad \rightarrow \quad f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$

Think – Pair – Share 4

An object of mass m is attached to a horizontal spring, stretched to a displacement A from equilibrium and released, undergoing harmonic oscillations on a frictionless surface with period T_0. The experiment is then repeated with a mass of $4m$. Choose the new period of oscillation.

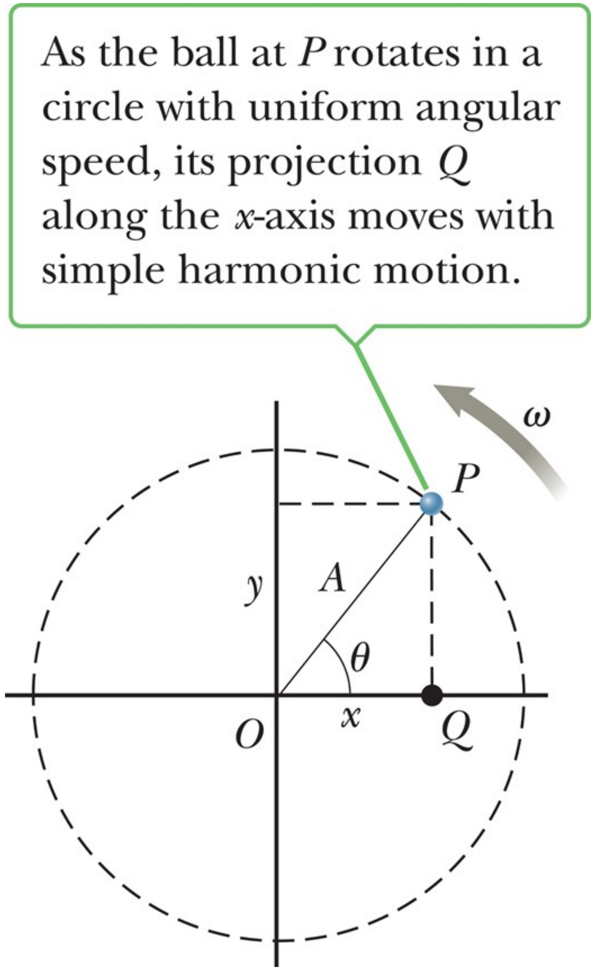
1. $2T_0$
2. T_0
3. $T_0/2$
4. $T_0/4$

Think – Pair – Share 5

An object of mass m is attached to a horizontal spring, stretched to a displacement A from equilibrium and released, undergoing harmonic oscillations on a frictionless surface. The experiment is then repeated with a mass of $4m$. Compared to the original total mechanical energy, the subsequent total mechanical energy of the object with mass $4m$ is

1. greater.
2. less.
3. equal.

Position, Velocity, and Acceleration as Functions of Time (1 of 4)



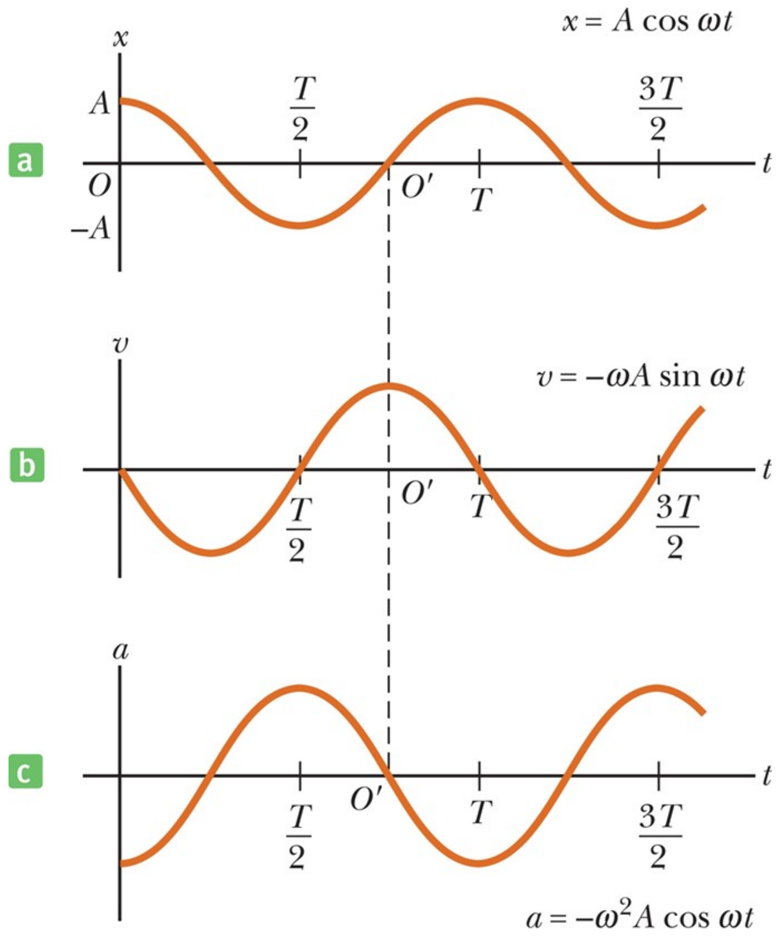
$$x = A\cos\theta$$

$$\theta = \omega t \quad \rightarrow \quad x = A\cos(\omega t)$$

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T} = 2\pi f$$

$$x = A\cos(2\pi f t)$$

Position, Velocity, and Acceleration as Functions of Time (2 of 4)



$$x = A\cos(2\pi ft)$$

$$v = \pm\sqrt{\frac{k}{m}(A^2 - x^2)} \quad \rightarrow$$

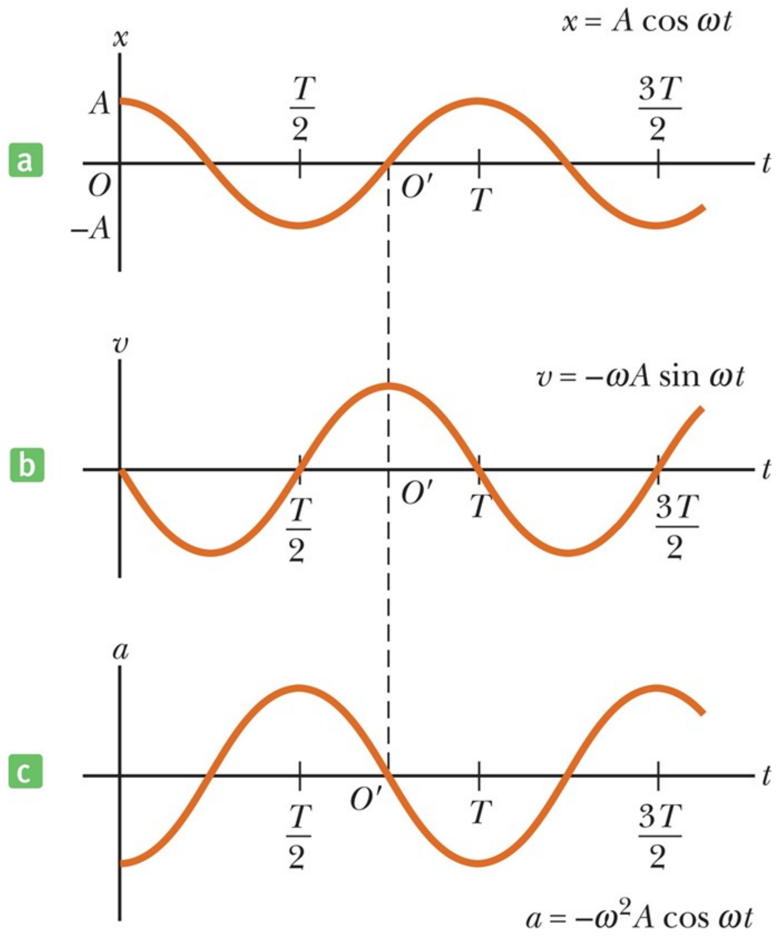
$$= \pm\sqrt{\frac{k}{m}(A^2 - A^2\cos^2(2\pi ft))}$$

Using:

$$1 - \cos^2\theta = \sin^2\theta \quad \text{and} \quad \omega = \sqrt{k/m}$$

$$\rightarrow \quad v = -A\,\omega\sin(2\pi ft)$$

Position, Velocity, and Acceleration as Functions of Time (3 of 4)

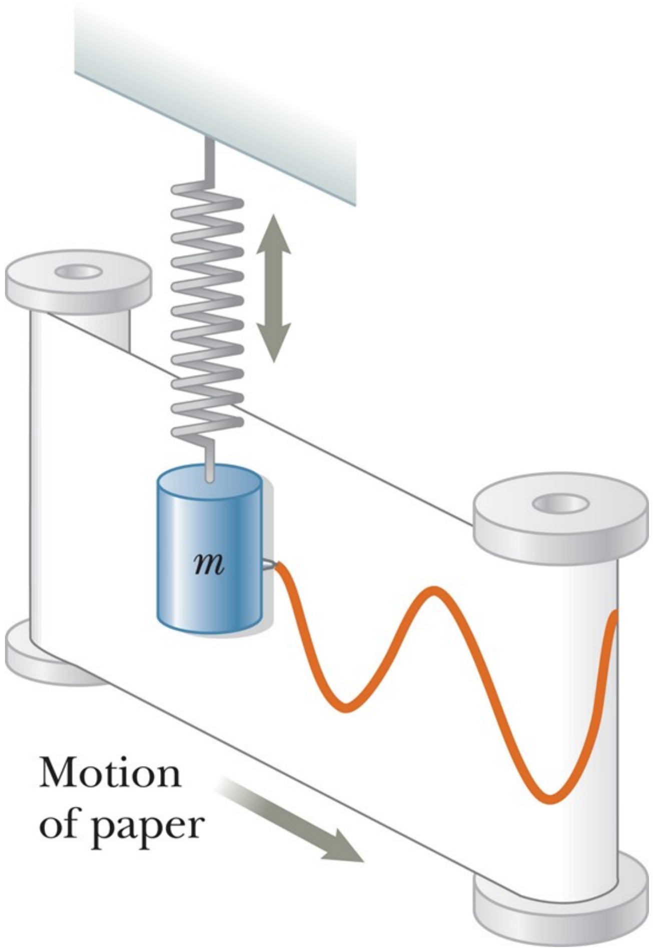


$$x = A\cos(2\pi f t)$$

$$a = -\frac{k}{m}x$$

$$a = -A\omega^2\cos(2\pi f t)$$

Position, Velocity, and Acceleration as Functions of Time (4 of 4)

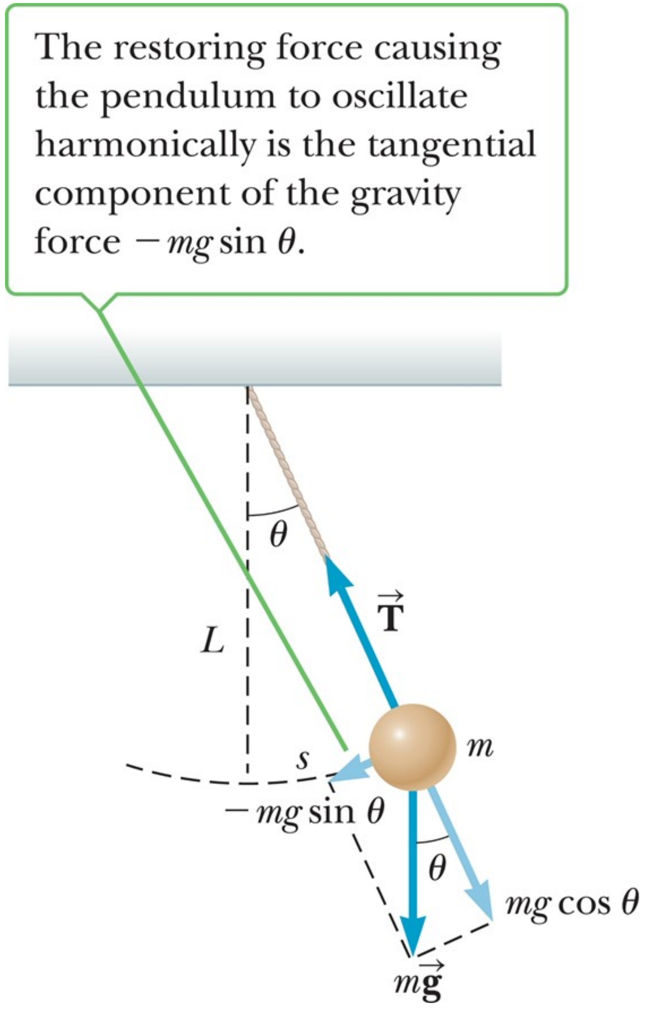


Think – Pair – Share 6

If the amplitude of a system moving in simple harmonic motion is doubled, which of the following quantities *doesn't* change?

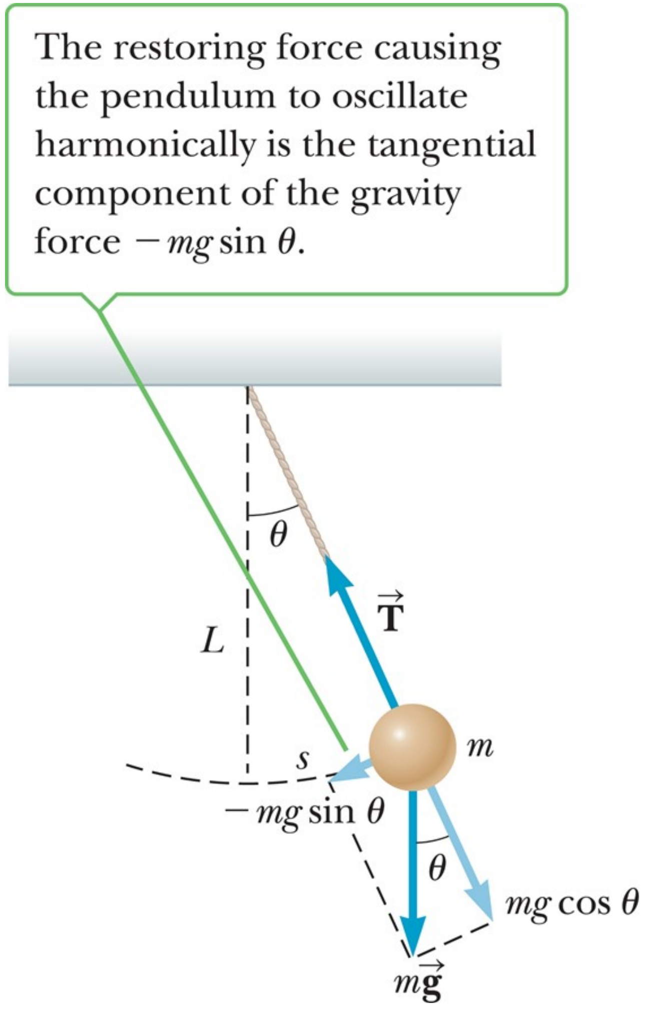
1. total energy
2. maximum speed
3. maximum acceleration
4. period

Motion of a Pendulum (1 of 5)



$$F_t = -mg\sin\theta \quad \rightarrow \quad s = L\theta$$
$$= -mg\sin\left(\frac{s}{L}\right)$$

Motion of a Pendulum (2 of 5)



for small angles:

$\sin\theta \approx \theta$

$\theta = 10.0^{\circ} = 0.175 \text{ rad}$

$\sin 10^{\circ} = 0.174$

$$F_t = -mg\sin\theta \approx -mg\theta$$

$$F_t = -\left(\frac{mg}{L}\right)s$$

Motion of a Pendulum (3 of 5)

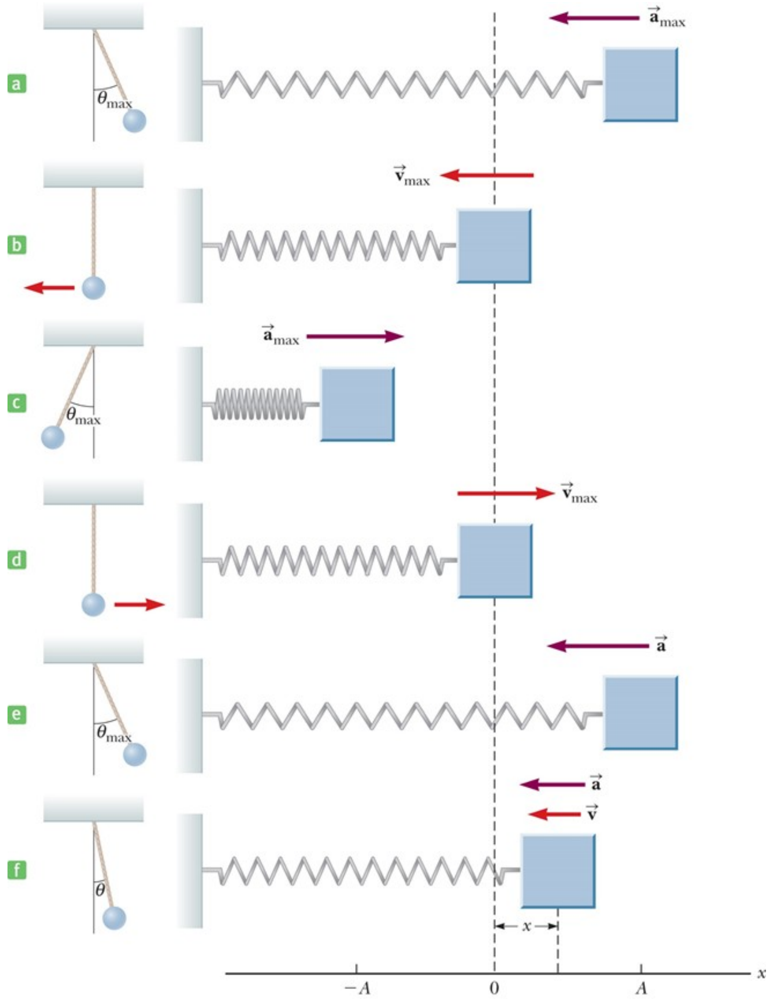
$$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{mg/L}{m}} = \sqrt{\frac{g}{L}}$$

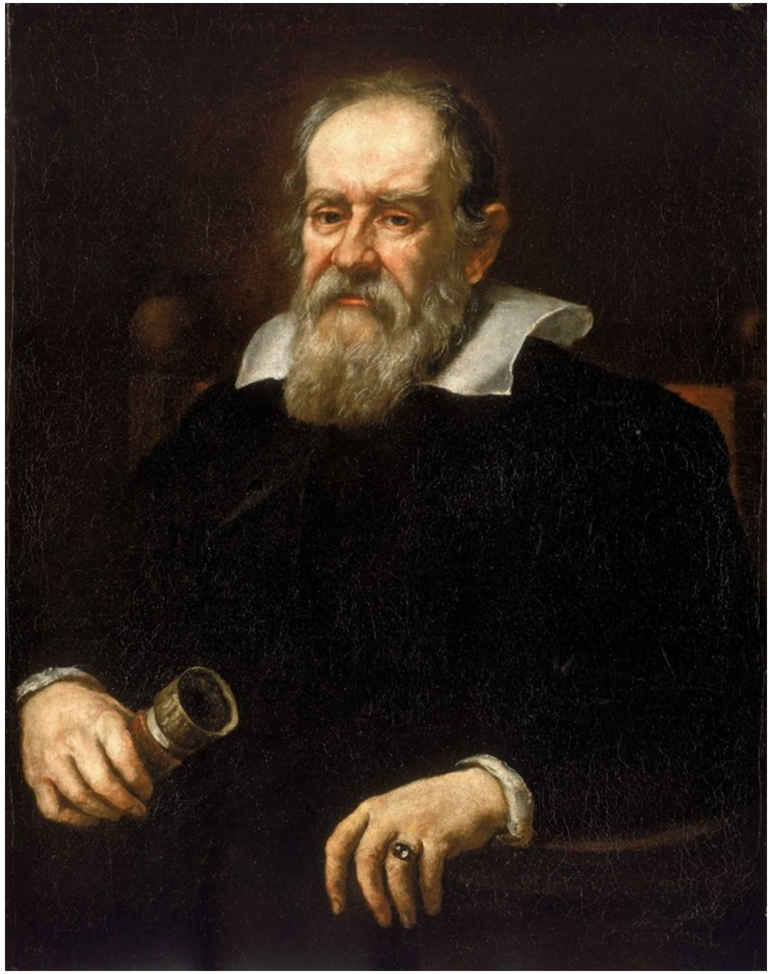
$$x = A\cos(\omega t) \quad \rightarrow \quad x = A\cos\left(\sqrt{\frac{g}{L}}\,t\right)$$

$$\omega = 2\pi f \quad \rightarrow \quad T = 2\pi\sqrt{\frac{L}{g}}$$

Motion of a Pendulum (4 of 5)



Motion of a Pendulum (5 of 5)



Think – Pair – Share 7

A simple pendulum is suspended from the ceiling of a stationary elevator, and the period is measured. If the elevator moves with constant velocity, the period

1. increases.
2. decreases.
3. remains the same.

Think – Pair – Share 8

A simple pendulum is suspended from the ceiling of a stationary elevator, and the period is measured. If the accelerates upward, the period

1. increases.
2. decreases.
3. remains the same.

Think – Pair – Share 9

A pendulum clock depends on the period of a pendulum to keep correct time. Suppose a pendulum clock is keeping correct time and then Dennis the Menace slides the bob of the pendulum downward on the oscillating rod. The clock then runs

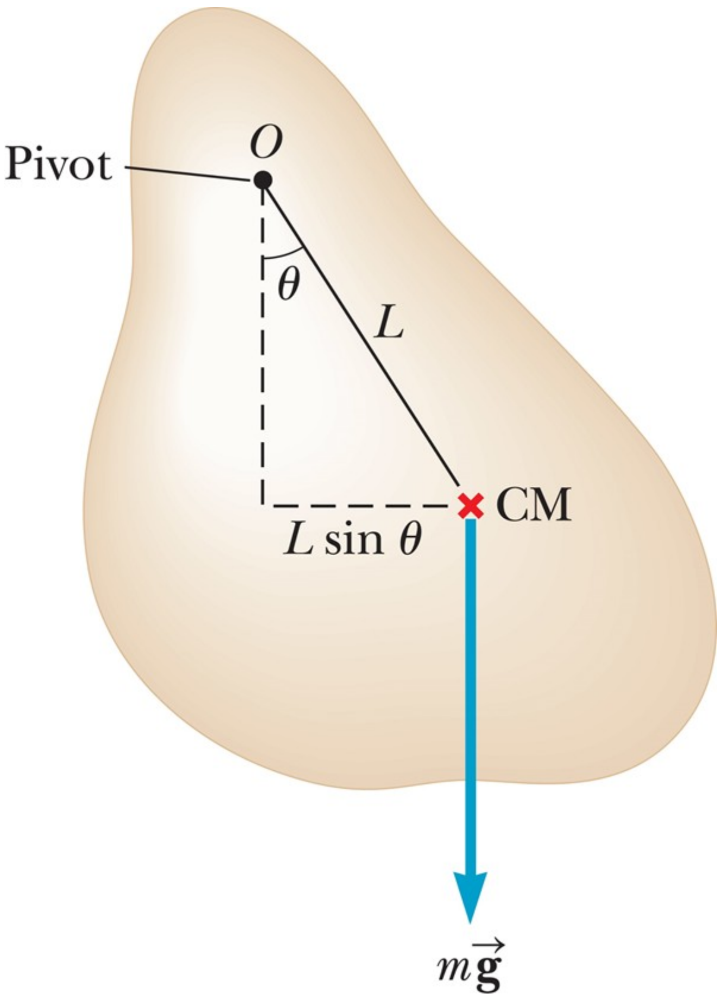
1. slow.
2. fast.
3. correctly.

Think – Pair – Share 10

The period of a simple pendulum is measured to be T on the Earth. If the same pendulum were set in motion on the Moon, would its period be

1. less than T
2. greater than T
3. equal to T

The Physical Pendulum



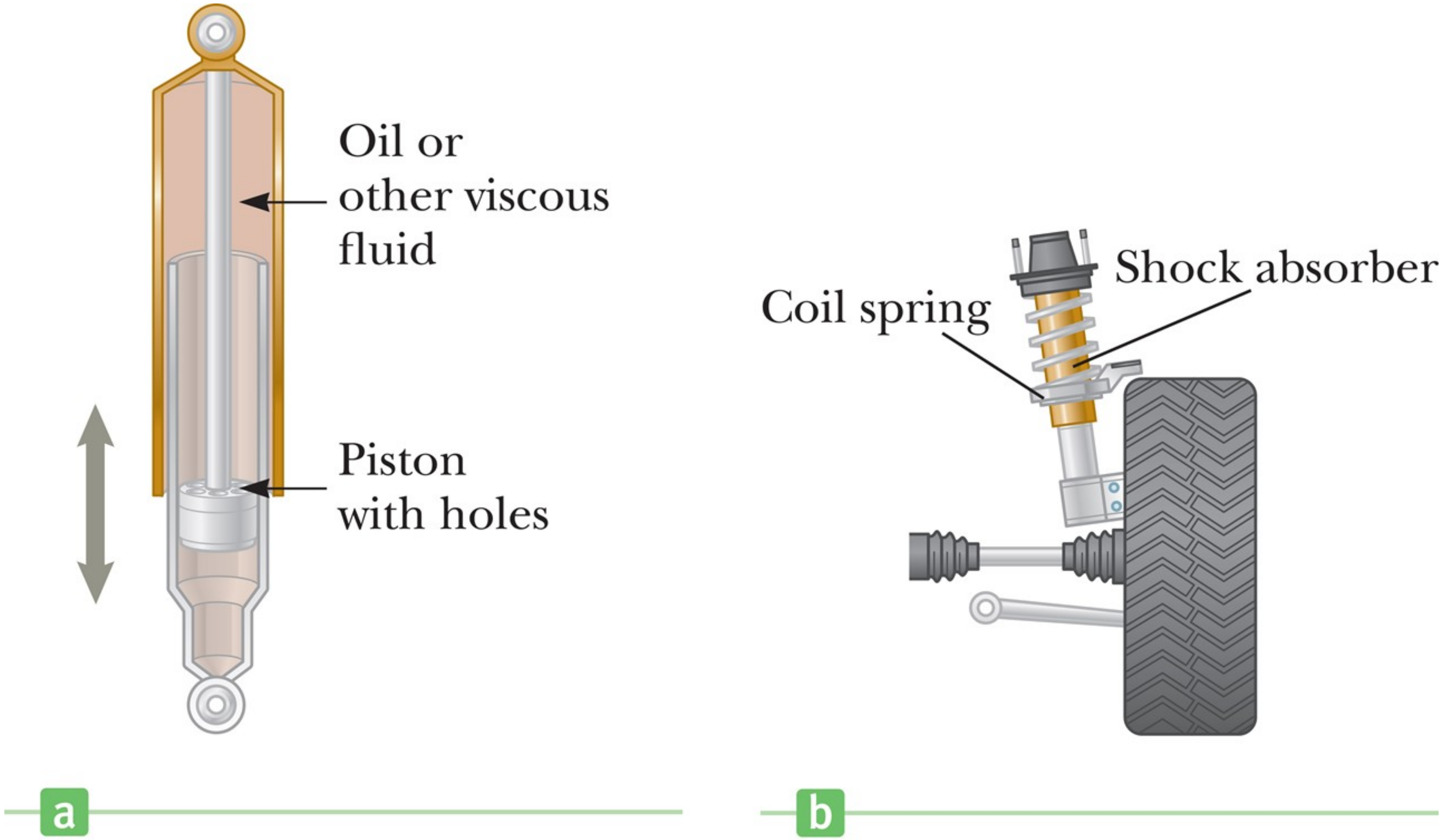
$$T = 2\pi\sqrt{\frac{I}{mgL}}$$

Simple pendulum:

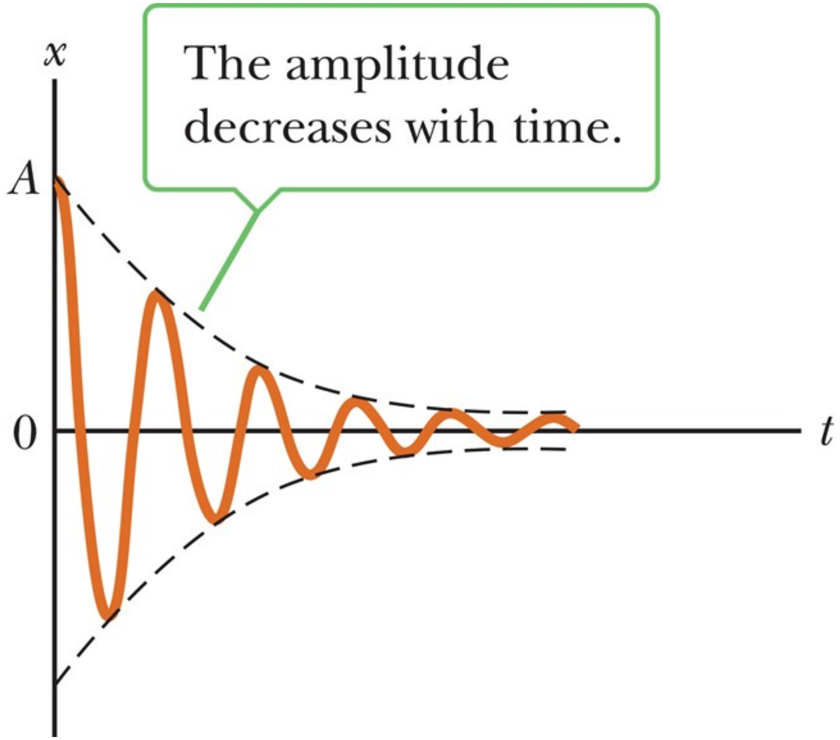
$$I = mL^2$$

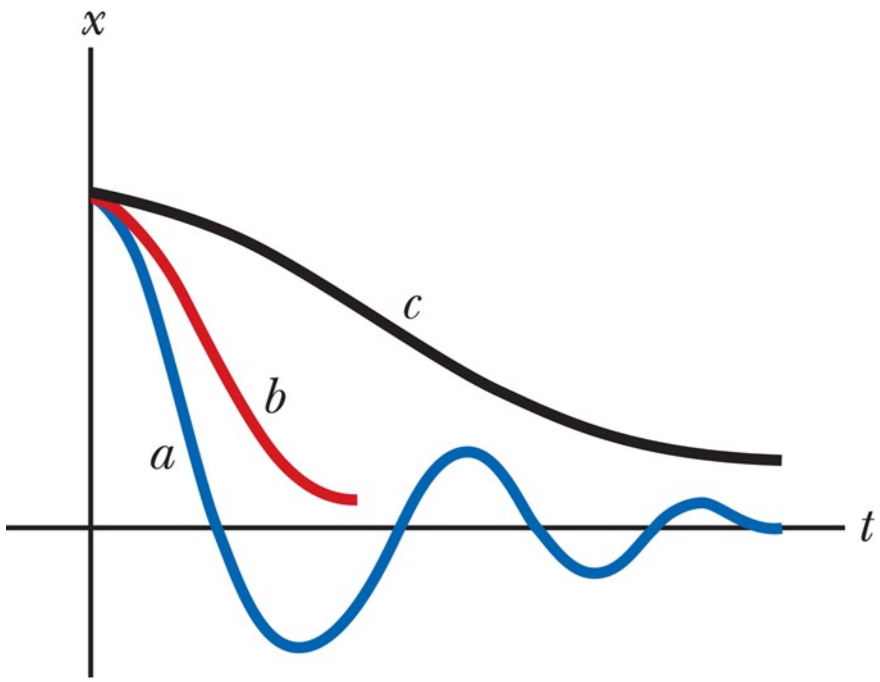
$$T = 2\pi\sqrt{\frac{mL^2}{mgL}} = 2\pi\sqrt{\frac{L}{g}}$$

Damped Oscillations (1 of 2)



Damped Oscillations (2 of 2)



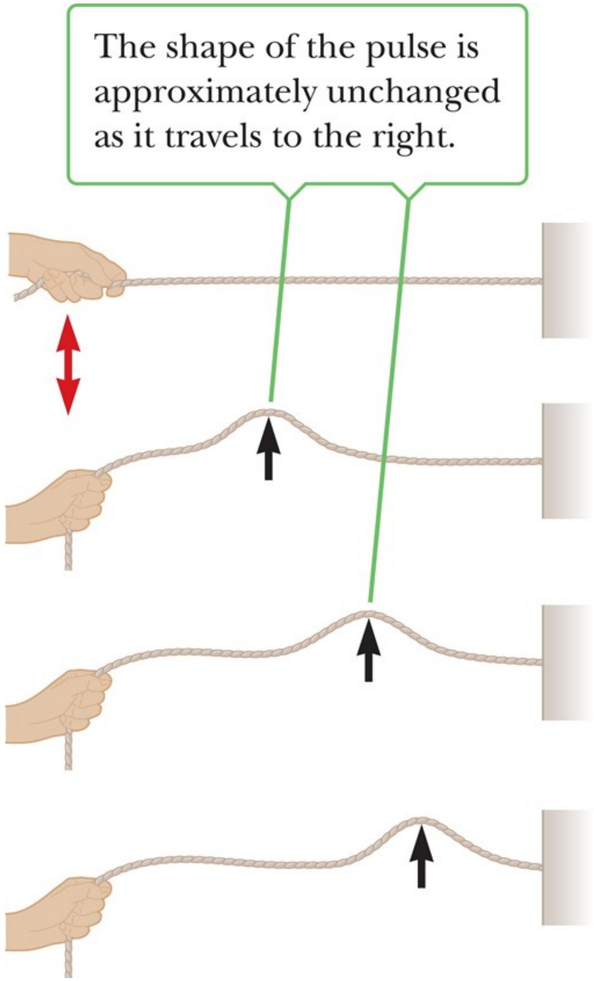


What Is a Wave? (1 of 2)

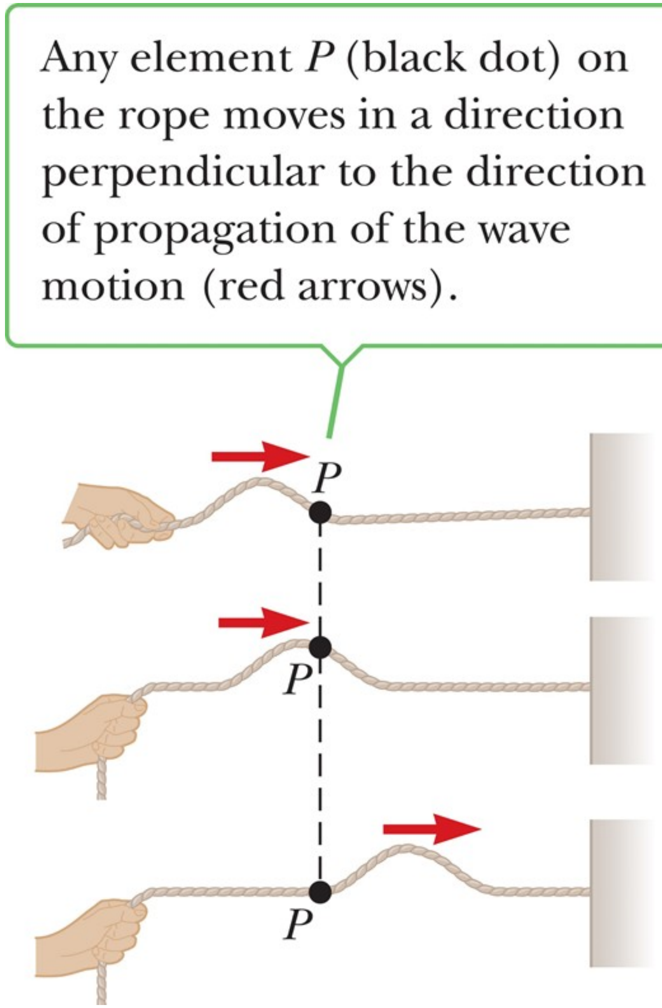
What Is a Wave? (2 of 2)

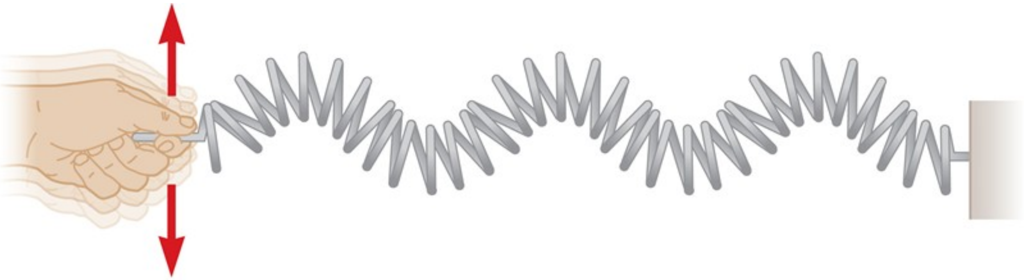
1. source of disturbance
2. a medium
3. a physical connection or mechanism

Types of Waves (1 of 4)



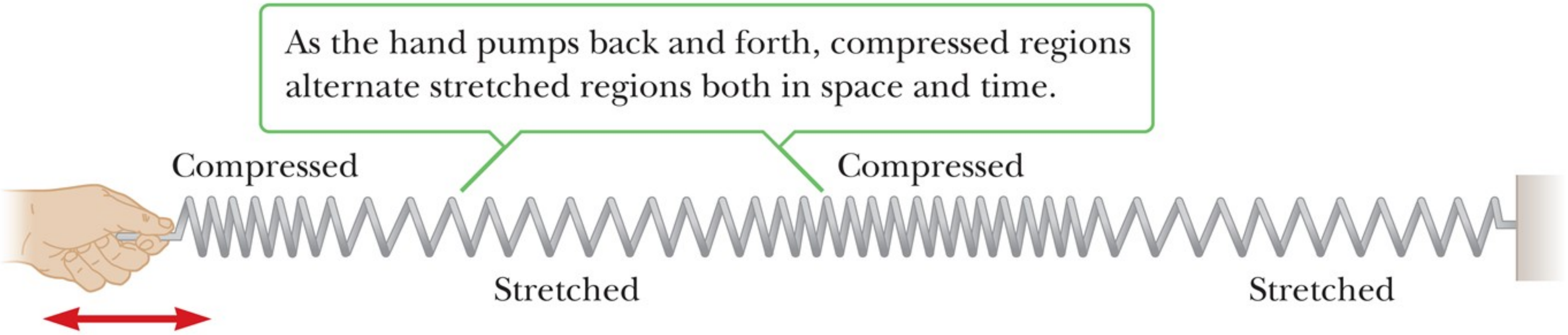
Types of Waves (2 of 4)





Transverse wave

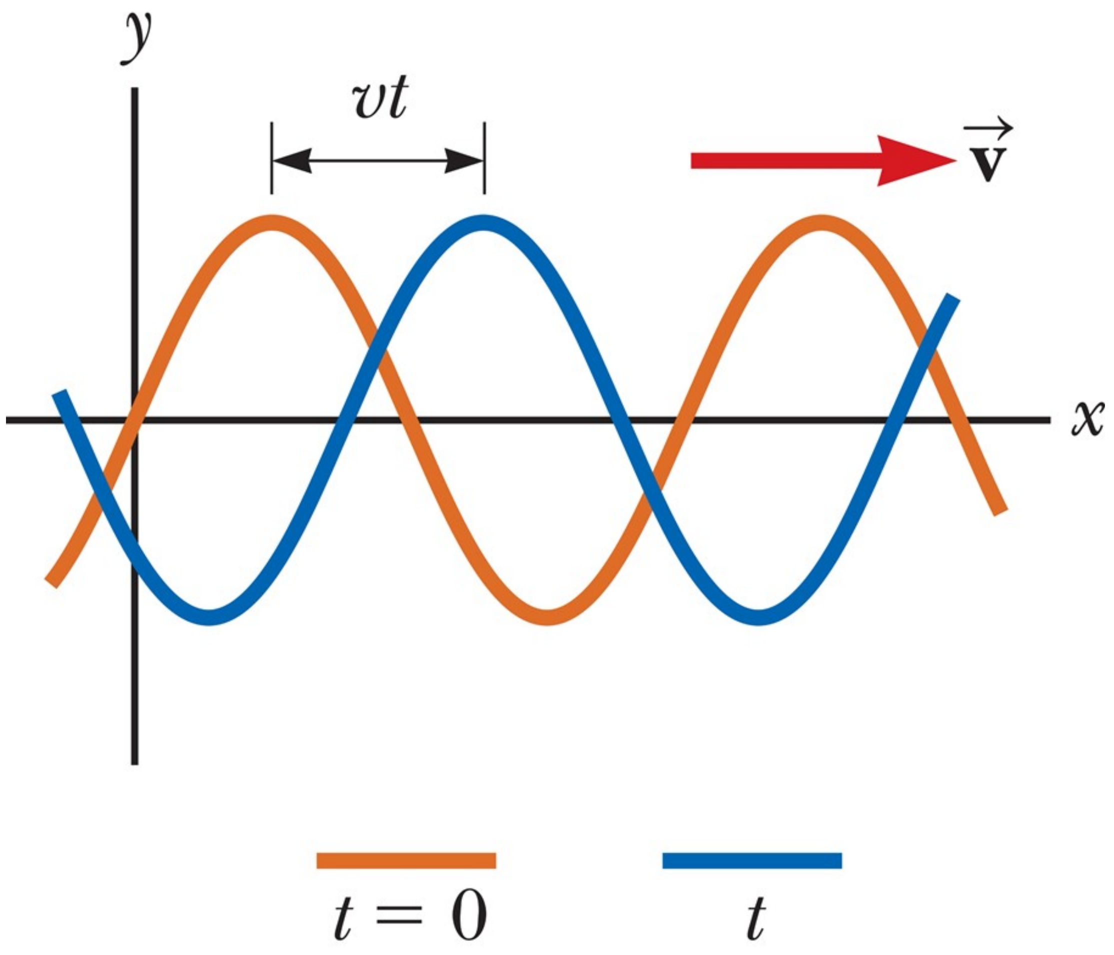
Types of Waves (3 of 4)



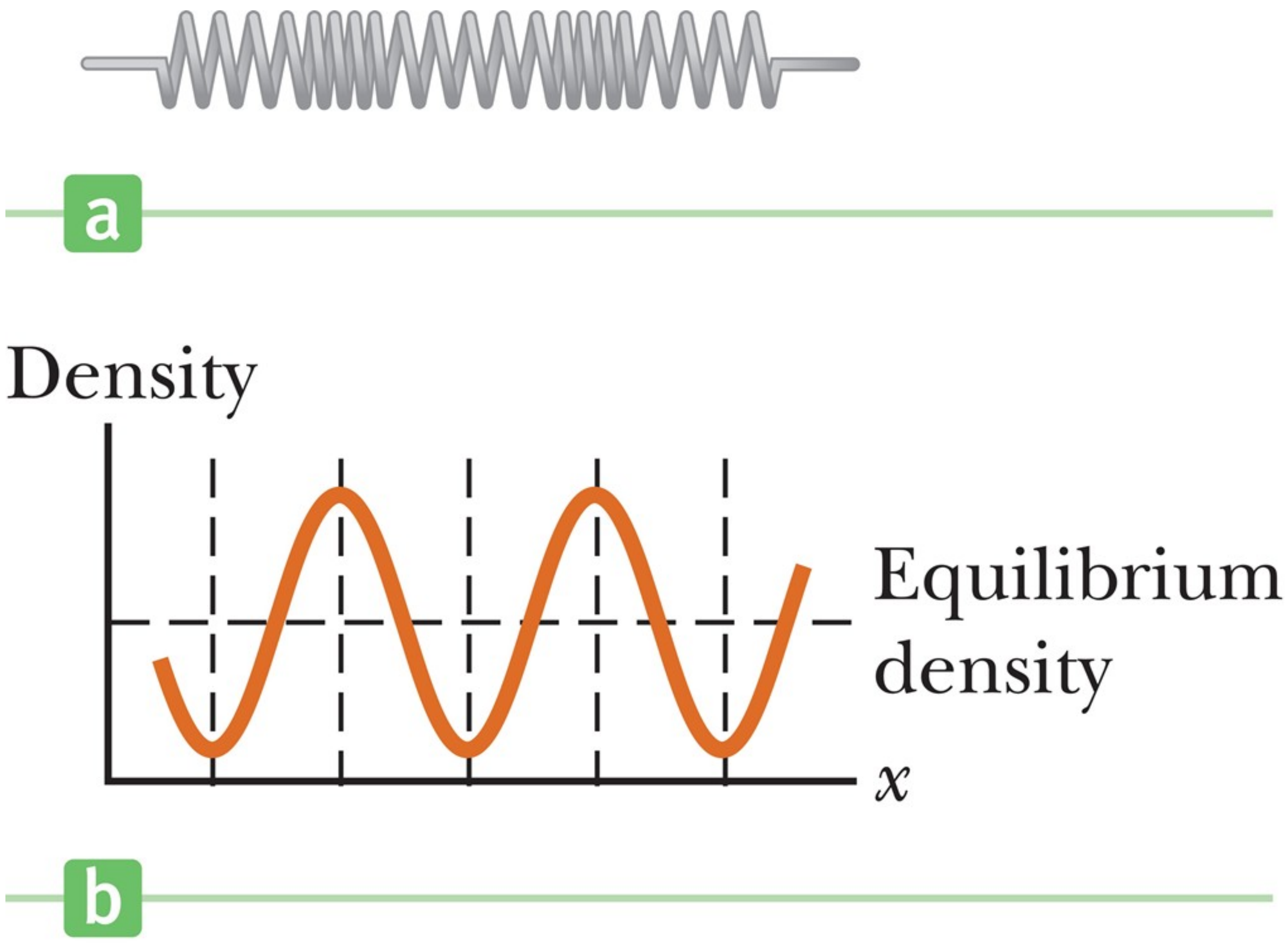
Longitudinal wave

Types of Waves (4 of 4)

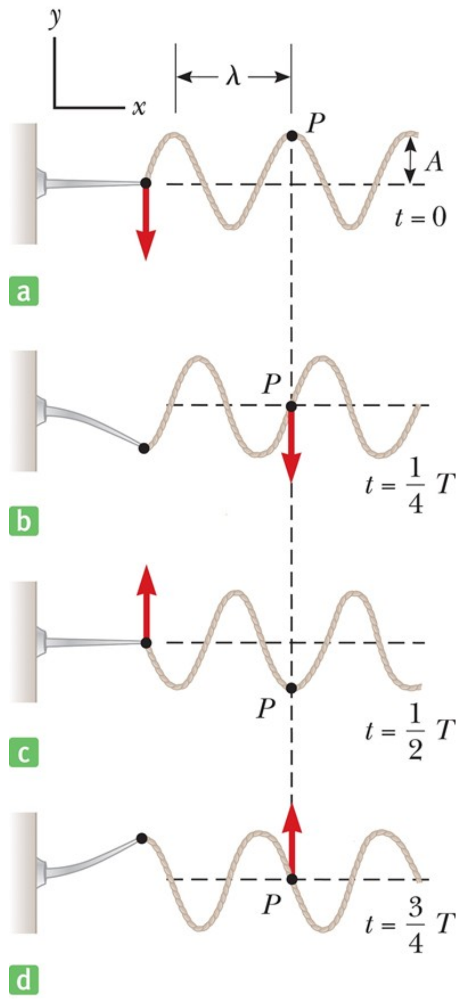
Picture of a Wave (1 of 2)



Picture of a Wave (2 of 2)



Frequency, Amplitude, and Wavelength (1 of 2)



Frequency, Amplitude, and Wavelength (2 of 2)

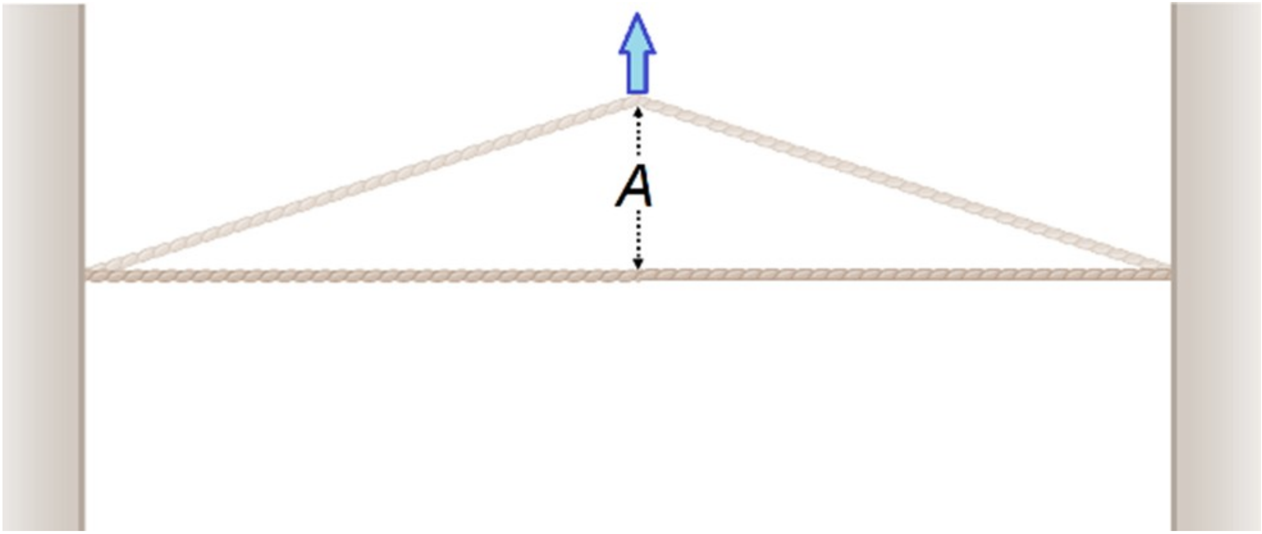
$$v = \frac{\Delta x}{\Delta t}$$

$$\Delta x \to \lambda \quad \text{and} \quad \Delta t \to T$$

$$v = \frac{\lambda}{T}$$

$$v = \lambda f$$

The Speed of Waves on Strings



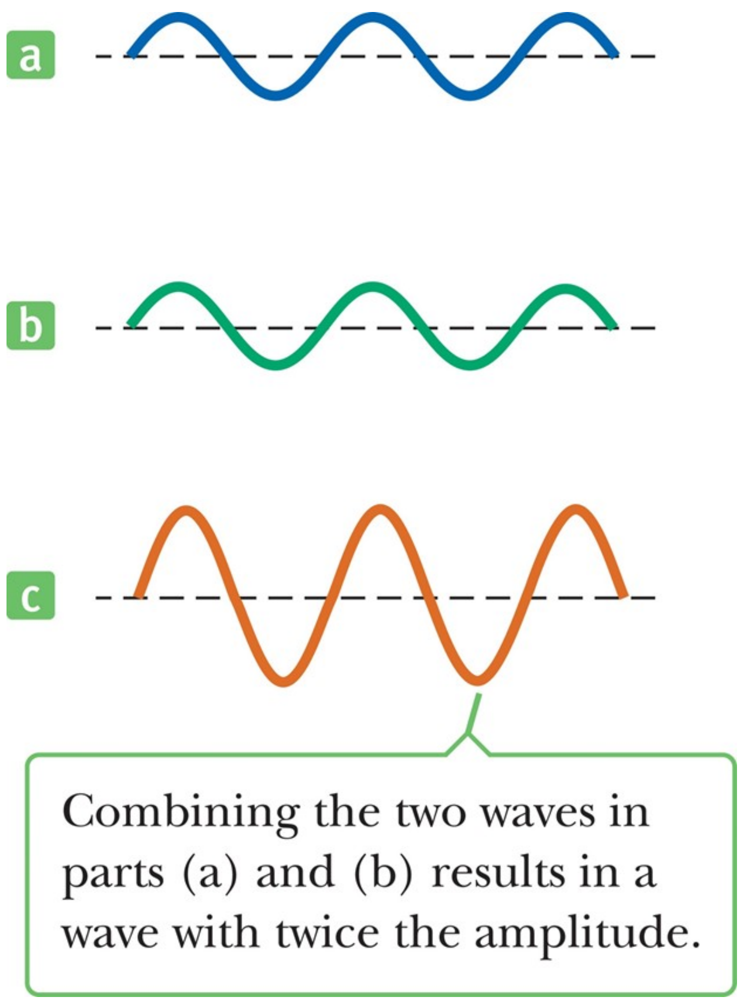
$$v = \frac{\lambda}{T}$$

$$v = \sqrt{\frac{F}{\mu}}$$

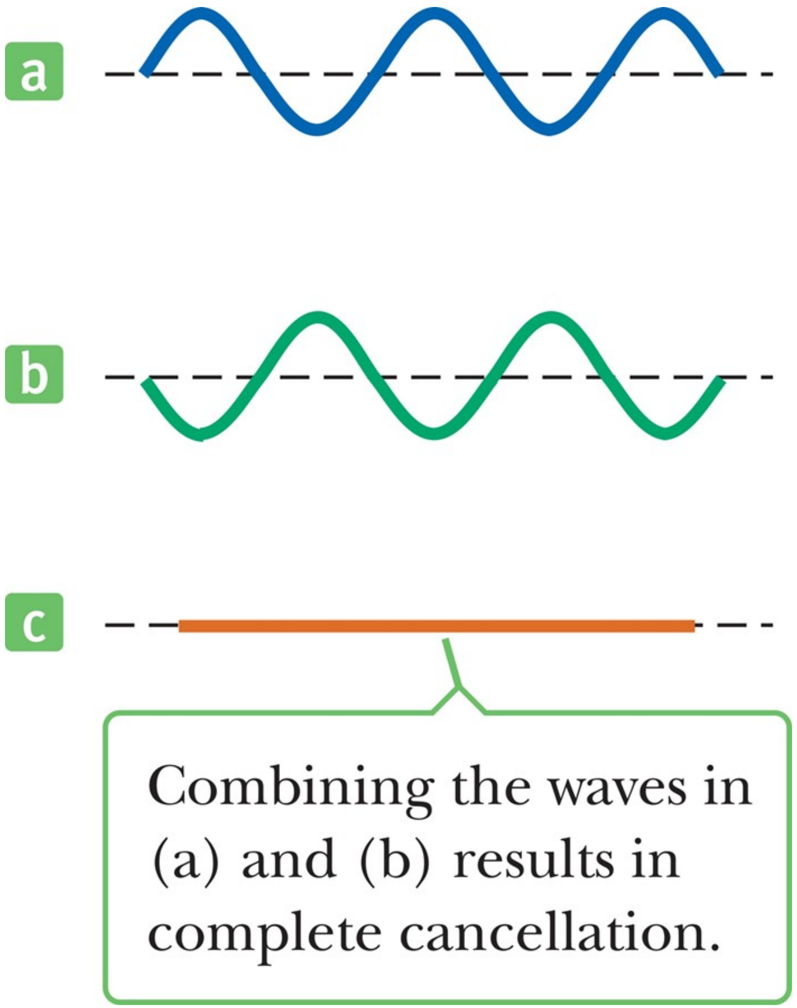
Interference of Waves (1 of 5)

Martin Dohrn/SPL/Science Source

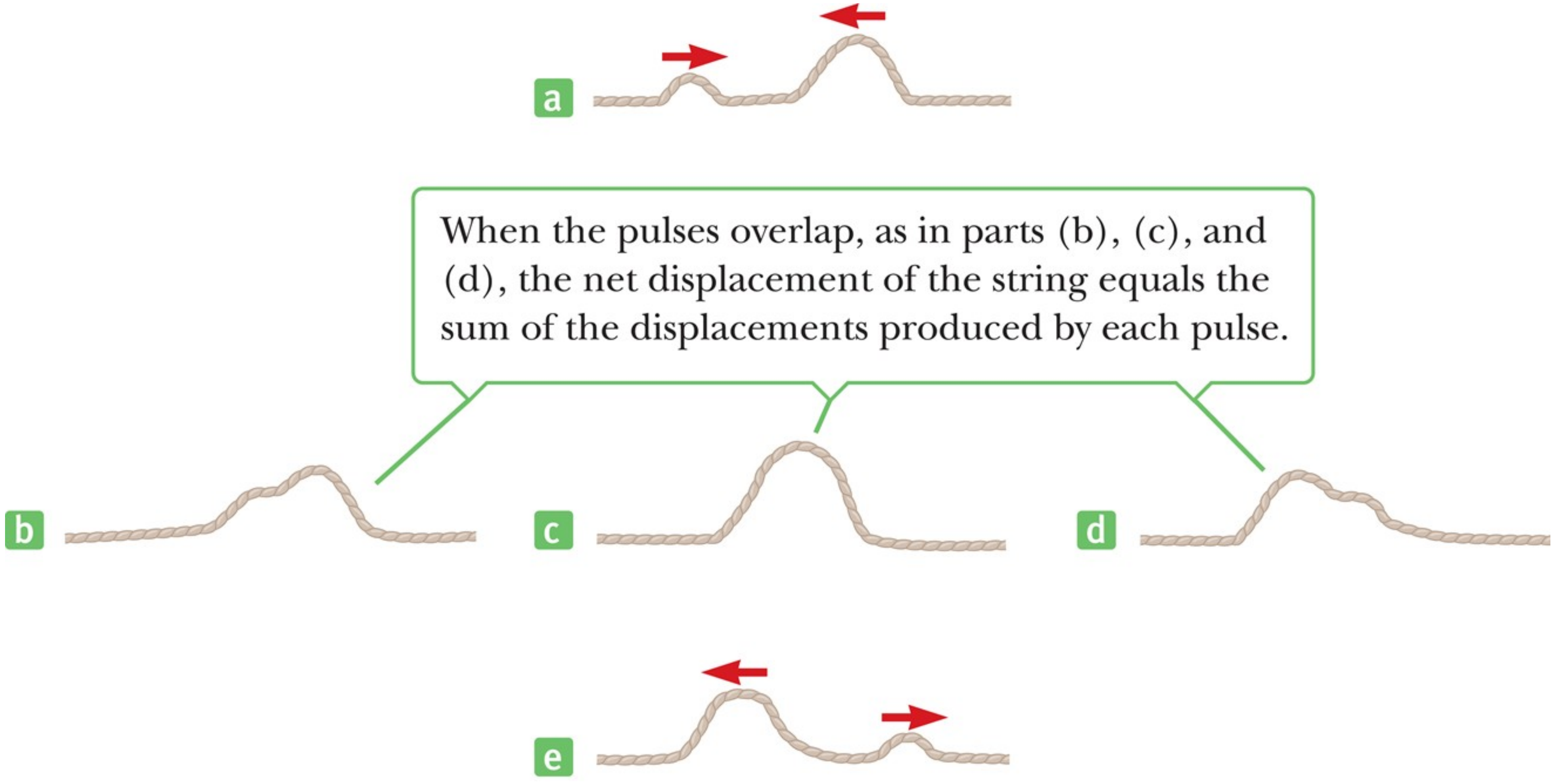
Interference of Waves (2 of 5)



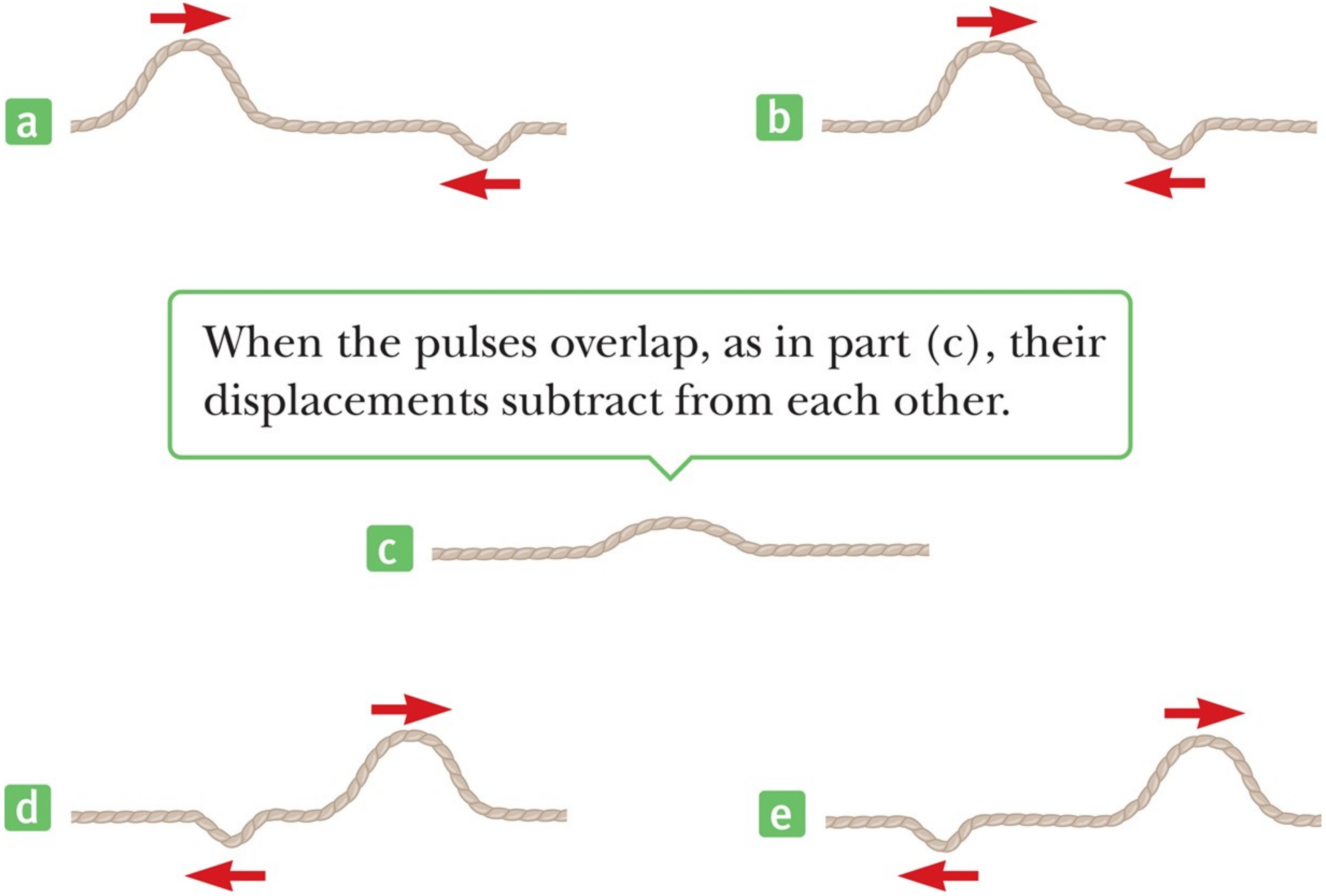
Interference of Waves (3 of 5)



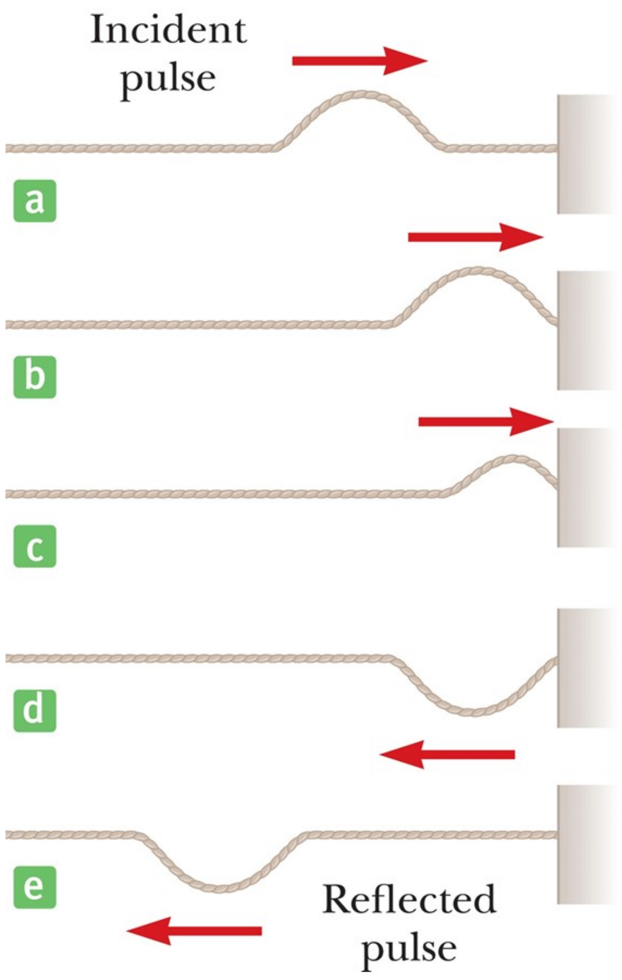
Interference of Waves (4 of 5)



Interference of Waves (5 of 5)



Reflection of Waves (1 of 2)



Reflection of Waves (2 of 2)